

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) reveal how small input changes flip a model’s prediction, yet most existing methods modify far more features than necessary, degrading their clarity and actionability. We introduce *COLA*, a model- and generator-agnostic post-hoc framework that refines any given CE by computing a coupling via optimal transport (OT) between factual and counterfactual sets and using it to drive a Shapley-based attribution (*p*-SHAP) that selects a genuinely minimal set of edits while fully preserving the target effect. We establish firm theoretical guarantees: OT provably minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and, under mild conditions, refined counterfactuals are guaranteed never to move farther from the factuals than the originals. Empirically, across four datasets, twelve models, and five CE generators, COLA consistently attains the same target effects using only 26–45% of the original feature edits, and on a tractable benchmark it matches the exact optimum. Together, these results establish COLA as a principled and broadly applicable solution to sparse counterfactual explanation.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) is indispensable for making artificial intelligence systems transparent and trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods, such as Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify precisely how much each input feature contributes to a machine learning (ML) model’s output, simplifying complex models by pinpointing the most influential features. In a healthcare model, for example, Shapley values isolate key factors such as age and medical history, directly aiding clinicians in interpreting the model’s decisions (Ter-Minassian et al., 2023; Nohara et al., 2022). Counterfactual Explanations (CE) (Wachter et al., 2017; Guidotti, 2022) instead show how small changes in input features lead to different outcomes. Although hundreds of CE algorithms have been proposed to date (Guidotti, 2022; Verma et al., 2020), no single CE algorithm can serve all use cases: each is deliberately tailored to its own scenario, goal, and task. One CE algorithm may seek a single counterfactual instance for each factual instance, while another treats multiple instances as a group and seeks one or more counterfactual instances per factual observation. Some algorithms operate on the entire dataset, identifying global CE that indicate the direction in which factual instances must move to achieve the desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024). Others view the group of factual instances as a distribution and seek a counterfactual distribution that preserves the shape of the factual distribution (You et al., 2025) at comparable cost. Some CE algorithms further assume differentiable models, whereas others are built specifically for tree-based or ensemble models. This diversity is precisely why a unifying, assumption-light framework is needed.

Problem Description and Challenges We tackle a general and comprehensive problem in this paper, building directly on the extensive foundations established in the literature (see Appendix A). Both FA and CE are vital for making AI models interpretable and accountable, yet each is incomplete on its own: FA alone fails to provide tangible, actionable steps, and CE alone typically prescribes unnecessary feature changes that are impractical to enact. We therefore address a general and

comprehensive problem that neither approach solves in isolation, and we seek to answer the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
Ⓢ	⌨	Ⓜ	Ⓢ	⌨	Ⓜ	Ⓢ	⌨	Ⓜ	Ⓜ
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform aims to increase user registrations. The platform has collected data on user interactions, such as the amount of money spent (Ⓢ), the number of clicks (⌨), and whether the user has registered (Ⓜ). In the original data ($\mathbf{x}$), no users are registered. Action plans $\mathbf{z}'$ and $\mathbf{z}''$ both adjust user characteristics to reach the desired outcome ($\mathbf{y}^*$) of full registration and both achieve the same half counterfactual effect, yet $\mathbf{z}''$ accomplishes this with strictly fewer modifications than $\mathbf{z}'$ —exactly the kind of edit reduction our framework delivers.

Three major challenges must be overcome to address this problem, and our framework confronts each of them head-on. First, no single CE algorithm can universally meet all needs, since the problem is inherently task-specific; a solution must therefore operate on top of any generator rather than replace it. Second, the approach must avoid any strong assumption about the model—such as differentiability or special structure—to remain applicable across the full range of models. Third, FA such as feature importance is actively misleading when its scores are incoherent with the changes required for the counterfactual effect. Put plainly, it is ineffective to perform FA independently of CE to select the most important features to change. We show in Result II of Section 6 that this decoupling produces counterproductive feature modifications (i.e., actions demanded of users/stakeholders), because the features deemed important in general simply do not align with the specific pathways that achieve the desired counterfactual outcomes.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that refines CE across arbitrary models and CE generators by using an optimal transport (OT)–induced coupling between factual and counterfactual sets to guide Shapley attributions. This yields p -SHAP, which subsumes the commonly used Shapley methods as special cases under appropriate couplings and delivers a clean, modular interface for attribution and edit selection. We back these claims with theoretical guarantees: OT provably minimizes an upper bound on the W_1 divergence, and under mild conditions refined counterfactuals are guaranteed to be no farther from the factuals than the originals. Empirically, across four datasets, twelve models, and five CE methods, COLA with p -SHAP attains the same target effects using only 26–45% of the original feature edits, and on a tractable benchmark it matches the exact optimum.

To the best of our knowledge, this is the first method to systematically solve the problem in Figure 1 without committing to any particular CE algorithm or model.

2 PROBLEM FORMULATION

We formally formulate the problem described in Figure 1. Denote $f : \mathbb{R}^d \rightarrow \mathbb{R}$ as a black-box ML model. Denote by $\mathbf{x}$ any observed (factual) data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$). Let $\mathbf{y}^*$ be the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The optimization is to look for a (group of) counterfactual data instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) subject to a maximum number of allowed feature changes, C , to achieve model output(s) as close as possible to $\mathbf{y}^*$. Let D denote a divergence

function that measures the dissimilarity between two entities. The problem is below.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) and the constraint equation (1b) formalize the typical CE optimization: $\mathbf{z}$ must drive $f(\mathbf{z})$ close to $\mathbf{y}^*$ while staying close to $\mathbf{x}$, so that $\mathbf{z}$ serves as a contrastive reference explaining why $f(\mathbf{x})$ fails to reach $\mathbf{y}^*$. We deliberately impose no restriction on the divergence D , keeping the formulation fully general; D may be Euclidean distance, OT, maximum mean discrepancy (MMD), or the mean or median difference of model outcomes. Constraints equation (1c)–equation (1e) then turn this into an *action-constrained* CE optimization. Alongside the counterfactual data $\mathbf{z}$, we jointly optimize an indicator variable $\mathbf{c}$ such that z_{ik} is permitted to change if and only if $c_{ik} = 1$, with at most C changes allowed by equation (1c). Constraints equation (1d) and equation (1e) enforce this exactly: when $c_{ik} = 0$, z_{ik} equals x_{ik} and nothing changes at (i, k) ; when $c_{ik} = 1$, the sufficiently large constant M grants z_{ik} full freedom to change.

To solve equation (1), we harness FA to pinpoint the most influential features and thereby fix the modification indicator variable $\mathbf{c}$. The next section reviews the commonly used Shapley value methods for FA which, together with our newly proposed method, are integrated into our algorithmic framework COLA to produce the refined counterfactual $\mathbf{z}$. The problem is genuinely hard: it remains computationally intractable even when $d = 1$ for linear models (see Appendix B), which underscores the need for the efficient attribution-driven approach we develop here.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews the commonly used Shapley value methods for FA that, together with our newly proposed method, are integrated into our algorithmic framework COLA. We first recall the concept of the Shapley value in game theory and then examine the various Shapley methods used for FA; (Sundararajan & Najmi, 2020) provides a comprehensive overview.

In cooperative game theory, a coalitional game is characterized by a finite set of players (here, features) $\mathcal{F} = \{1, 2, \dots, d\}$ and a characteristic function v . This function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ maps each subset $S \subseteq \mathcal{F}$ to a real number $v(S)$, the total payoff achievable by the members of S through cooperation, with $v(\emptyset) = 0$. The Shapley value, defined below in terms of v , is the canonical solution concept in this framework, providing the unique equitable distribution of the grand coalition’s payoff $v(\mathcal{F})$ among the individual players.

Prior to defining the Shapley value, denote $\Delta(k, S)$ the incremental value produced by a player (feature) k to a coalition S , aligning with the concept of marginal contribution. This is defined as the additional payoff realized by the coalition due to the inclusion of the player k , i.e.,

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subseteq \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k averages the normalized marginal contributions $\Delta(k, S)$ of feature k across all subsets S of the feature set excluding k , with normalization given by the inverse binomial coefficient $\binom{d-1}{|S|}$ and d the total number of features. Summing over all subsets and dividing by d yields the fair, axiomatically grounded distribution of payoffs among players, faithfully reflecting each feature’s contribution within the full collection of features.

For an arbitrary data point $\mathbf{x}_i$, we use $\mathbf{x}_{i,S}$ to represent its part containing only the features in S , and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ the other remaining part. The same rule applies for other notations as well. Below, we introduce commonly used Shapley value methods.

Baseline Shapley (B-SHAP). This approach relies on a specific baseline data point $\mathbf{r}_j$ as the reference for any $\mathbf{x}_i$, with set function

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j, \mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where a feature’s absence is modeled by its value in the reference baseline data point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This method rests on the strong assumption of an exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This variant of B-SHAP is the one implicitly used in (Lundberg & Lee, 2017) and in the widely adopted “SHAP” library, given by

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ is the background distribution that is commonly selected to be the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This extends RB-SHAP by taking the distribution $\mathcal{D}$ to be the counterfactual distribution defined on each individual factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method, adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021), has shown clear advantages for contrastive FA, but it still assumes a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We unify equation (4) through equation (6) under a single formulation by integrating an algorithm A_{Prob} that returns their joint probability.¹ Our p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, whereas prior methods lean on random baselines, p -SHAP resolves the alignment problem directly by identifying the optimal coupling via OT. This recasts feature attribution—no longer a mere prediction decomposition, but a transport problem that provably minimizes the cost of explanation.

Note that A_{Prob} is independent of the CE algorithm and depends only on the generated counterfactual $\mathbf{r}$. By depending solely on these fixed components, p -SHAP guarantees consistent FA that is immune to the variability of different CE generation processes—a decisive difference from CF-SHAP. Contrary to common expectations, we show that OT is more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022), established in Result II of Section 6.

A central focus of this paper is the OT problem (equivalently, the 2-Wasserstein divergence) defined below, whose transportation plan $\mathbf{p}_{\text{OT}}$ we use directly as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ in p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions (all admissible transport plans). The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularization with coefficient $\varepsilon \geq 0$; setting $\varepsilon > 0$ substantially accelerates the computation of OT.

¹First, p -SHAP reduces exactly to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP reduces to RB-SHAP in equation (5) when A_{Prob} is independent of CE and instead uses an arbitrary distribution $\mathcal{D}$. Third, p -SHAP reduces to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

Theoretical Aspects of p -SHAP OT minimizes the total feature modification cost (moving $\mathbf{x}$ towards $\mathbf{r}$) under the alignment it induces between factual $\mathbf{x}$ and counterfactual $\mathbf{r}$, which is exactly our objective of achieving the counterfactual outcomes at minimal cost. We make this connection precise under the mild assumption that the predictive model f is Lipschitz continuous. Theorem 4.1 below (proof in Appendix C) proves that the transportation plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$: the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant of f times the square root of the minimized expected cost of changing $\mathbf{x}$ towards $\mathbf{r}$, i.e. $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where the p_{ij} ($j = 1, 2, \dots, m$) quantify how the feature values of $\mathbf{x}_i$ should be adjusted towards those of one or multiple $\mathbf{r}_j$. In practice, this means the OT plan $\mathbf{p}_{\text{OT}}$ prescribes exactly how to shift the feature values of $\mathbf{x}$ towards those of $\mathbf{r}$ at minimal expected modification cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$. Among all admissible modification plans ($\mathbf{p} \in \Pi$), $\mathbf{p}_{\text{OT}}$ achieves the smallest possible cost and therefore delivers the tightest upper bound on the violation of the counterfactual effect $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Theorem 4.1 bounds the transport cost and thereby furnishes a tractable convex proxy for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem. By concentrating attribution mass on the most efficient paths, the OT coupling reliably guides the greedy selection in Algorithm 1 to discard non-essential features. Moreover, p -SHAP is provably correct in attributing causal behavior to the modifications of the features, as stated in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Furthermore, p -SHAP inherits the desirable axioms of B-SHAP and RB-SHAP, which establishes it as a principled and effective tool for feature attribution. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the Shapley axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, stated in Algorithm 1 below, solves equation (1) and is built on four categories of algorithms. We explain Algorithm 1 in details below, along with an illustration in Figure 2.

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE algorithm A_{CE} takes the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ as input, and returns a counterfactual $\mathbf{r}$ that stays close to $\mathbf{x}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$. Any CE generator can be plugged in here.

Line 2 (Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$). The algorithm A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ as input and outputs a matrix representing the joint distribution over all n and m data points in $\mathbf{x}$ and $\mathbf{r}$, respectively. This joint distribution $\mathbf{p}$ encodes the alignment (matching) between factual and counterfactual rows, and Line 5 uses it to compute the values that compose $\mathbf{z}$. As discussed in Section 3, when the row alignment is unknown, A_{Prob} should be based on OT to compute the joint distribution that attains the smallest OT distance between $\mathbf{x}$ and $\mathbf{r}$; when a reliable alignment is available, A_{Prob} should adopt the joint distribution that faithfully reflects the alignment between the rows of $\mathbf{x}$ and $\mathbf{r}$.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

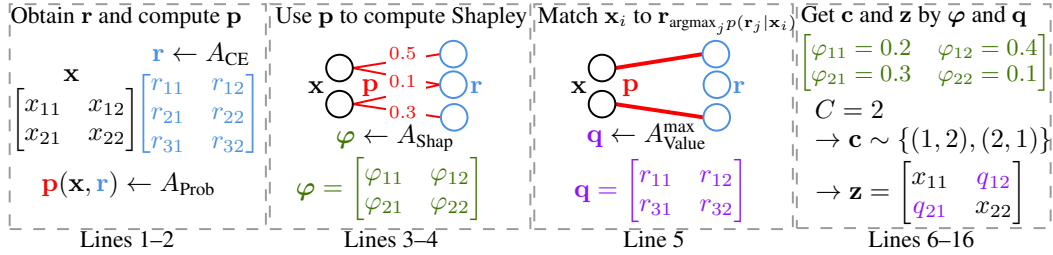


Figure 2: [An illustration of COLA] This figure shows how COLA gets $\mathbf{c}$ and $\mathbf{z}$ for equation (1). We use $A_{\text{Value}}^{\max}$ for illustration in line 5 due to its simplicity. In lines 6–16, we assume $C = 2$, and the sampling yields exactly two positions for modifications according to the probability matrix φ .

Lines 3–4 (*p*-SHAP FA). We apply equation (7) to compute the Shapley values for $\mathbf{x}$. The joint distribution $\mathbf{p}$ is used here to align each row of $\mathbf{x}$ with its corresponding counterfactual rows of $\mathbf{r}$, so that the selected rows of $\mathbf{r}$ serve as the most representative contrastive reference for the row in $\mathbf{x}$. This alignment has a decisive numerical impact on our contrastive FA. The resulting Shapley-value matrix of $\mathbf{x}$ is then taken element-wise in absolute value and normalized to sum to one, yielding the matrix φ that constitutes our FA.

Line 5 (*Computing feature values*). The algorithm A_{Value} takes the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$ as input. For each row i of $\mathbf{x}$, A_{Value} selects one or multiple rows of $\mathbf{r}$ to serve as references for the changes to $\mathbf{x}$, and returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ in which each element q_{ik} is the counterfactual candidate for x_{ik} ($\forall i, k$). We introduce $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$ for the single-row and multiple-row cases, respectively.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ selects the row of $\mathbf{r}$ with the highest probability.

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i), k} \text{ and } \tau(i) = \arg \max_{j=1, 2, \dots, m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ computes q_{ik} as a convex combination as a weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Since the non-negative matrix φ is normalized to sum to one, we treat it directly as a selection policy over the positions in $\mathbf{x}$ to replace (i.e. $c_{ik} = 1$), exactly as line 7 does. For every

i and k with $c_{ik} = 1$, x_{ik} is set to q_{ik} ; the modified matrix is returned as $\mathbf{z}$, and together with $\mathbf{c}$ it constitutes the optimized solution of equation (1).

By Theorem 4.1, COLA minimizes the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by modifying $\mathbf{z}$ according to the feature attributions, which identify precisely the most important features to adjust to achieve the desired counterfactual effect. The theorem below (proof in Appendix E) proves that the refined $\mathbf{z}$ produced by COLA satisfies the constraint equation (1b) in the typical scenario where $n = m$, under the Frobenius norm. Table 3 confirms empirically that this conclusion holds far more broadly.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ be the algorithm complexity of A_{CE} . For algorithm A_{Shap} , consider using weighted linear regression to estimate Shapley values, and denote by M_{Shap} the number of sampled subsets. The complexity of COLA with respect to n, m, d , and the regularization parameter ε of entropic OT is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used. See Appendix F for more details.

6 NUMERICAL RESULTS

This section evaluates COLA on the problem in equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is the counterfactual obtained from a CE method A_{CE} . We adopt four divergence functions: OT measures the distance between entire distributions; MMD measures the divergence between the means of two distributions in a high-dimensional feature space; and absolute mean difference (MeanD) and absolute median difference (MedianD) measure the divergence in mean and median, respectively. The results establish three claims: I) COLA delivers substantial modification minimality. II) p -SHAP decisively outperforms other Shapley methods on modification minimality. III) COLA attains near-optimal performance.

Experiment Setup We conduct experiments over 4 binary-classification datasets, 5 CE algorithms designed for diverse goals, and 12 classifiers, shown in Table 1, where each combination of dataset, A_{CE} algorithm, and model defines an “experiment scenario”. This broad sweep stress-tests COLA across a wide range of settings.

We briefly introduce the A_{CE} methods in Table 1. DiCE and KNN are instance-based CE methods, yielding counterfactual(s) for each factual instance. ARoS and GlobeCE are group-based CE methods, finding a collection of counterfactual instances for the whole factual data as a group. Discount treats the factual instances as an empirical distribution and seeks a counterfactual distribution close to it. This selection spans the full spectrum of CE paradigms.

Table 2 defines the 6 methods, where CF- p_{OT} is our proposed p -SHAP and the others are baselines. Each is placed in numerous experiment scenarios from Table 1 and benchmarked comprehensively. Each method is fixed by a combination of the three algorithms A_{Prob} , A_{Value} , and A_{Shap} . For example, the first row RB- p_{Uni} uses the uniform distribution as A_{Prob} to compute $\mathbf{p}$, feeds $\mathbf{p}$ into equation (9) or equation (10) of A_{Value} to compute $\mathbf{q}$, and sets A_{Shap} to RB-SHAP.

Table 1: [Experiment Scenarios Setup] Four datasets benchmark five CE algorithms across 12 models. A “scenario” is a combination of dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The table defines the 6 methods compared, colored to match Figure 3. Each method is placed in an experiment scenario, as defined in Table 1, and benchmarked.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

These methods are deliberately designed to isolate each factor through ablation. First, RB- p_{Uni} differs from CF- p_{Uni} only in that the latter uses CE information while the former does not. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all use CE in FA but with different joint distributions, and we show that the OT-computed distribution outperforms the alternatives by a wide margin. Third, to confirm that OT helps precisely because it informs A_{Shap} with the factual-counterfactual alignment—and not through any confound—we compare CF- p_{OT} , which uses this alignment, against RB- p_{OT} , which does not. Finally, CF- p_{Ect} is the special case in which each counterfactual originates from a known source, used as an exact factual-counterfactual alignment, so that CF-SHAP coincides with B-SHAP.

Result I: COLA achieves drastic action reduction at negligible loss in counterfactual effect

In Table 3, we require $\mathbf{z}$ such that $f(\mathbf{z})$ reaches 80% or 100% of the counterfactual effect of $f(\mathbf{r})^2$. COLA attains this goal while requiring far fewer actions in $\mathbf{z}$ (feature modifications) than the original CE $\mathbf{r}$: it reaches the 80% counterfactual effect using only 13%–25% of the feature changes (measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$). Decisively, COLA with p -SHAP (CF- p_{OT}) is the only method that reaches the full 100% counterfactual effect, and it does so with just 26%–45% of the feature changes in $\mathbf{z}$ relative to the original $\mathbf{r}$. p -SHAP thus delivers the best action minimality by a clear margin, as we analyze in detail below.

Result II: p -SHAP outperforms all other Shapley methods in achieving the counterfactual effect

Figure 3 evaluates the Shapley methods of equation (4)–equation (7), with the number of allowed feature changes C on the x-axis and $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1) on the y-axis. Three findings stand out. First, RB- p_{Uni} and RB- p_{OT} perform markedly worse than the rest, proving that CE information is essential for attributing features towards modification minimality in CE. Second, CF- p_{OT} clearly beats RB- p_{OT} , and since the two differ only in A_{Shap} (see Table 2), this pins the gain squarely on the effective factual-counterfactual alignment that OT supplies to p -SHAP, ruling out confounding factors. Third, p -SHAP substantially outperforms CF- p_{Uni} and CF- p_{Rnd} , demonstrating the effectiveness of the OT joint distribution: merely using counterfactual information for FA (as CF- p_{Uni} and CF- p_{Rnd} do) is insufficient, and a proper alignment—which, as Figure 4 reveals, need not be the one defined by the exact counterfactual generation mechanism—is indispensable.

²By definition $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, a 100% counterfactual effect, since $\mathbf{y}^* = f(\mathbf{r})$. A counterfactual effect of 80% is defined by the proportion of divergence reduced by the refined CE $\mathbf{z}$, i.e. $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

Table 3: [COLA for Modification Minimality] Number of modified features in $\mathbf{z}$ for each method when $f(\mathbf{z})$ reaches the 80% counterfactual effect of $f(\mathbf{r})$. Each entry is averaged over 4 randomly selected scenarios from Table 1, with 100 runs per scenario; best values are in bold. The symbol “-” indicates that the target counterfactual effect cannot be achieved at all.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

COLA with p -SHAP performs strongly across the experiment scenarios of Table 1, as Figure 3 and Table 3 both show, and Appendix H corroborates this with an extensive set of additional results. In short, p -SHAP outperforms every other Shapley method for sparsity in CE.

Result III: COLA matches the MILP optimum on tractable benchmarks This result shows that p -SHAP eliminates the influence of the CE generation process by replacing the CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between factual and counterfactual data, as shown in Figure 4. We benchmark CF- p_{Ect} within COLA, solving equation (1) with MeanD as the divergence D . Because CF- p_{Ect} relies on a known factual-counterfactual alignment, this isolates how effectively COLA exploits such an alignment. The theoretical optimum of COLA in this case is obtained by solving an MILP (Appendix G derives the formulation); since the MILP is computationally heavy, we solve it for *German Credit*.

CF- p_{Ect} attains near-optimal performance using DiCE. For DiCE and KNN the factual-counterfactual pairs are mutually independent, so COLA is demonstrably effective for instance-based CE even though our formulation in equation (1) is a strict generalization to group and distributional CE. Strikingly, CF- p_{OT} sometimes outperforms CF- p_{Ect} , because it identifies the key features to modify through a more theoretically grounded route, whereas CF- p_{Ect} relies on CE algorithm-dependent knowledge that carries no solid justification for its effectiveness in FA. A gap between CF- p_{Ect} , CF- p_{OT} , and the optimum remains for KNN, and finding the best alignment is a promising open direction.

7 CONCLUSIONS

This paper introduces COLA, a novel framework that refines CE through joint-distribution-informed Shapley values and guarantees that the refined CE preserves the counterfactual effect while requiring substantially fewer actions. Its core innovation—coupling factual and counterfactual sets via OT

³This knowledge, i.e. the exact alignment between factual and counterfactual, is available only from DiCE and KNN among all CE methods considered.

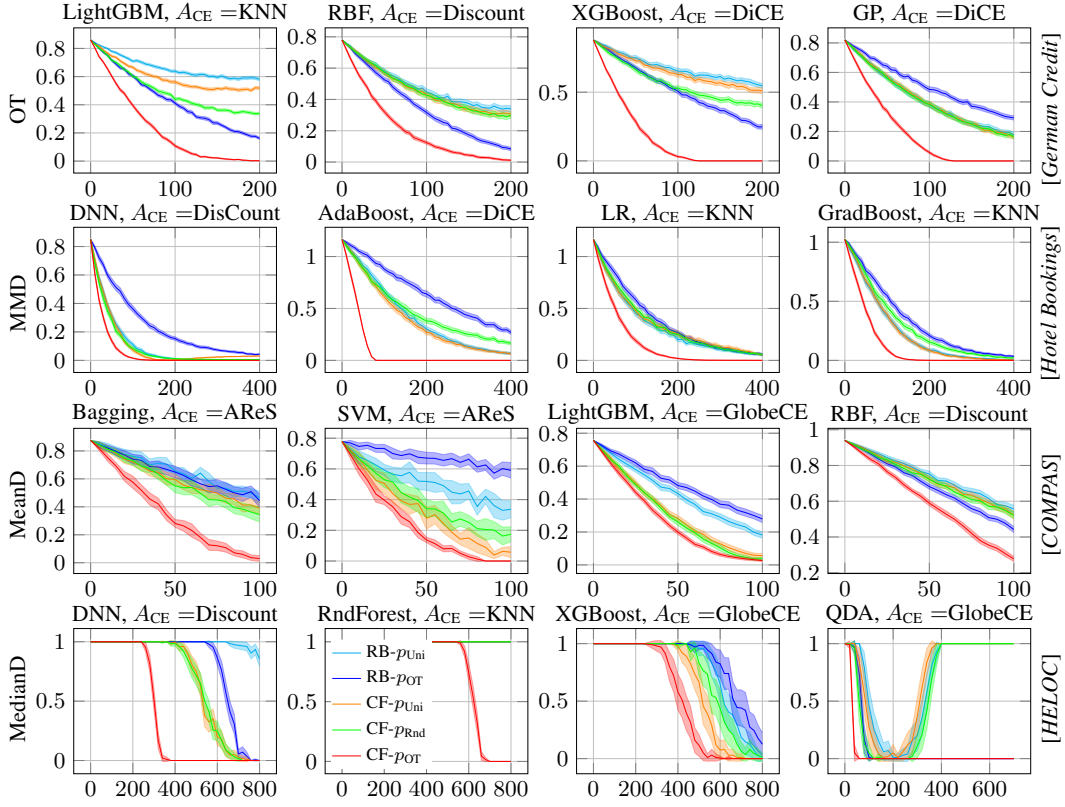


Figure 3: $D(f(z), y^*)$ vs. allowed actions C , over 100 runs; shaded bands are 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{Value}}^{\text{max}}$ for *German Credit* and *Hotel Bookings*. The legend in “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all plots. Across every panel, $\text{CF-}p_{\text{OT}}$ (p -SHAP) drives the divergence to the target fastest.

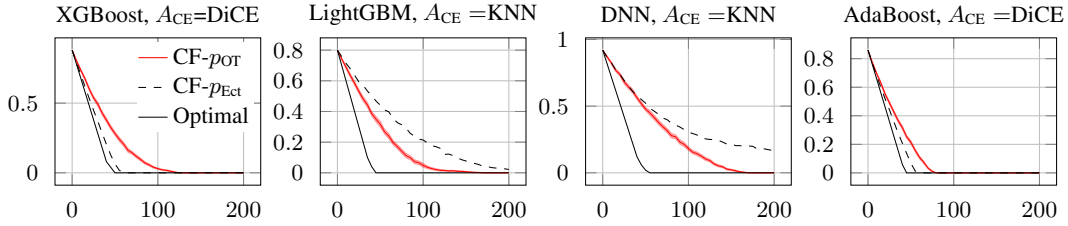


Figure 4: [*German Credit*] $D(f(z), y^*)$ vs. allowed actions C , with D being *MeanD*. COLA tracks the exact mixed integer linear programming (MILP) optimum closely, and $\text{CF-}p_{\text{OT}}$ at times even surpasses the known-alignment $\text{CF-}p_{\text{Ect}}$.

to supply Shapley attribution with a principled alignment—is backed by theoretical guarantees and confirmed across four datasets, twelve models, and five CE generators, where COLA consistently reaches the target effects with only 26–45% of the original edits and matches the exact optimum on a tractable benchmark. COLA is model- and generator-agnostic, and thus provides a practical, broadly applicable foundation for sparse counterfactual explanation; extending it to richer data modalities and to settings where a causal graph is available is a natural next step.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a method (COLA with p -SHAP) for refining CE. Our work is methodological and empirical; it does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We follow dataset licenses/terms and the common research practice of using these data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility/immutability constraints on features.

CE can surface biases in underlying models but do not by themselves guarantee fairness. Our method can reduce the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age) nor does it confer causal validity. We discourage use of CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should include domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact/benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional (not causal) and must not be interpreted as causal effect.

Risks include (i) misinterpretation of attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these by (a) stating non-causal scope and limitations in the paper, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing guidance for responsible use in documentation.

We provide an anonymized implementation, detailed settings, and seeds to facilitate replication; we report model types, hyperparameters, and compute budgets, and we avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We take reproducibility seriously and provide pointers to all necessary components. The COLA framework and p -SHAP attribution are specified with pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), with all theoretical assumptions and complete proofs deferred to the appendix (theorems and lemmas with line-by-line proofs).

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which also describes the datasets used and the evaluation protocol (including data splits and preprocessing). Baselines (models and CE generators) and coupling choices are enumerated in the experimental sections and ablations, with their configurations mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; we also report hardware and runtime in Appendix I to contextualize compute requirements, and provide ablation/sensitivity studies and the small-scale MILP benchmark setup in the appendix to support independent verification.

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APPENDICES

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A COMPARISON TO EXISTING APPROACHES

Albini et al. (2022) proposed CF-SHAP, which uses counterfactual data points as the background distribution for Shapley. However, it assumes the counterfactual data distribution is defined conditionally on each single instance, which presupposes a known probabilistic alignment between every factual and every counterfactual instance; similar assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). This assumption breaks down in the many scenarios that demand global explanations, and the counterfactual-distribution setup of (Albini et al., 2022) is in fact a special case of ours. Kwon & Zou (2022) proposed WeightedSHAP, weighting features rather than treating them as equal contributors; our method instead weights the contributions of data points and extends straightforwardly to weighting both rows and columns. Kommiya Mothilal et al. (2021) builds a framework combining FA and CE for explainability, but its CE-based FA inherits the same assumption as (Albini et al., 2022), making it difficult to generalize to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). Another relevant work, (Sharma et al.), models counterfactuals via a system’s causal structure and time-series predictors to fairly decompose a single observed change in a system-level metric into per-input contributions. Crucially, none of this prior work addresses the minimal-actions CE problem that is the focus of our paper.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) are the closest to ours. Karimi et al. (2021) minimize the cost of performing actions under the assumption of a known structural causal model (SCM), which is rarely available in practice. Kanamori et al. (2022) note that it remains open whether existing CE methods can solve that problem, and propose an MILP-solver-based approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Solving MILP, however, is costly and scales poorly—precisely the limitation our OT-driven, assumption-light framework overcomes.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below establishes that no scalable exact algorithm can solve the problem in general unless $\mathcal{P} = \mathcal{NP}$; the hardness stems not merely from the non-linearity of f but fundamentally from its combinatorial nature. This is exactly why an efficient attribution-driven approach such as COLA is warranted.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the goal is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error while having at most K non-zero

elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard due to the combinatorial nature of selecting the subset of variables to include in the model.

We will map this SR problem to our problem equation (1) with the following settings. Let the number of features be $d = 1$ and the number of instances be n (the same as the dimension of the SR problem). Let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ (the zero vector), and the target model output be $\mathbf{y}^* \in \mathbb{R}^m$ (as given in the SR problem). We define the model f as a linear function $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output, we define $D(f(\mathbf{z}), \mathbf{y}^*) = \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2$, i.e. the Euclidean distance squared. For the instances, we define $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. We set the maximum allowed feature changes C to be equal to k (the sparsity level from the SR problem). The large constant M can be any sufficiently large positive number, for example, $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

This means that if $c_i = 0$, then $z_i \leq 0$ and $z_i \geq 0$, so $z_i = 0$. If $c_i = 1$, then $z_i \in [-M, M]$. The constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, the variables c_i indicate whether the variable z_i is allowed to change ($c_i = 1$) or not ($c_i = 0$). The constraints enforce that $z_i = 0$ when $c_i = 0$, mirroring the sparsity constraint in the SR problem. The constraint $\sum_{k=1}^n c_i \leq k$ ensures that at most k features can change, matching the sparsity level. The objective function is identical to that of SR.

Since our problem formulation directly mirrors the SR problem with a cardinality constraint, which is known to be $\mathcal{NP}$ -hard, solving Problem equation (1) is at least as hard as solving the SR problem. Therefore, Problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, the model f is linear, and the divergence functions D are standard Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below provides that feature attributions are aligned with the expected costs of feature modifications, leading to action plans that are cost-efficient in achieving counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention is independent of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$. Consequently, $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ is exactly the combined effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$ —confirming that p -SHAP attributes effects in a causally meaningful way.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We analyze the computational complexity of COLA step by step, and show it is efficient.

First, we analyze A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} just constructs the matrix $\mathbf{p}$ with the prior knowledge, which takes $\mathcal{O}(n \times m)$. Otherwise, we consider solving OT to obtain $\mathbf{p}$, which, by the Sinkhorn–Knopp algorithm, takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Then we analyze A_{Shap} . For each subset, the model is evaluated on all n data points, leading to $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ involves replacing the values of certain features with their corresponding baseline values. This operation is $\mathcal{O}(m)$ because it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. If we assume that the reference values can be precomputed and accessed in constant time, then the complexity of incorporating these values can be considered as $\mathcal{O}(d)$. The number of M_{Shap} subsets results in M_{Shap} model evaluations. Combining the above steps, the complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\text{max}}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{max}}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\text{max}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA with respect to n , m , d , and the regularization parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

In this section, we derive a global-optimality benchmark for solving equation (1) with a known alignment between factual and counterfactual and D being MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is used, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ stays closer to $\mathbf{x}$ than $\mathbf{r}$, hence equation (1b) is dropped. The formulation of equation (1) then becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because $\text{CF-}p_{\text{Ect}}$ is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\max}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scaler variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented by a binary variable a_{iS} together with the scaler g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under the two constraints equation (14b) and equation (14c) is equivalent to minimizing the objective function of equation (13). The constraints in equation (14d) guarantees that each data point i is subject to one and only one feature modification plan a_{iS} for a specific S ($S \subseteq \mathcal{F}$). The constraint equation (14e) corresponds to equation (13b). Solving equation (14) yields the theoretical optimality of COLA using a known alignment between factual and counterfactual, demonstrated in Figure 4 in Section 6.

H EXTENDED NUMERICAL RESULTS

We analyze how the different Shapley methods affect FA, corresponding to lines 3–4 of COLA, via two controlled comparisons. First, $\text{RB-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Uni}}$ differ only in A_{Shap} ; across Figures 5–8, $\text{CF-}p_{\text{Uni}}$ consistently outperforms $\text{RB-}p_{\text{Uni}}$. Second, $\text{RB-}p_{\text{OT}}$ and $\text{CF-}p_{\text{OT}}$ also differ only in A_{Shap} , and the latter consistently outperforms the former. Both comparisons lead to the same firm conclusion: *RB-SHAP is unsuitable for FA in CE.*

Beyond FA, the equally important line 5 of COLA uses the joint probability $p(\mathbf{x}, \mathbf{r})$ to compose the matrix $\mathbf{q}$, which tells the factual x in which direction to change its features to move towards the target model outcome. Across Figures 5–8, CF- p_{OT} consistently outperforms every other method in all experiments. Although CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all provide a joint probability $\mathbf{p}$ when the exact factual-counterfactual alignment is unknown, their performance differs sharply—decisive evidence that simply knowing the CE (and its marginal distribution) is insufficient.

OT is exceptionally effective precisely when the alignment between factual and counterfactual instances is missing or inaccurate. Even when a CE algorithm explicitly matches each factual instance to a counterfactual, there is no guarantee that this known alignment optimizes performance, as Figure 4 in Section 6 confirms.

Importantly, p_{OT} need not be the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$; its role is to guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ jointly for both FA and CE. Furthermore, the QDA column in Figure 5 shows that OT-based methods remain stable while the others diverge markedly from the target. We stress that COLA is *not restricted to OT* as A_{Prob} : as Figure 4 indicates, even the best known $\mathbf{p}$ still leaves a non-negligible gap to global optimality, so designing an even stronger A_{Prob} is a compelling direction for future work.

I EXPERIMENTS REPRODUCIBILITY

The experiments run on a high-performance computing (HPC) cluster with four nodes (one per dataset) in parallel, each node equipped with two Intel Xeon Processor 2660v3 (10 core, 2.60GHz) and 128 GB memory. Each node takes roughly 5–10 hours depending on dataset size. The experiments also reproduce on a laptop, albeit at greater computational time than on the HPC cluster.

For all four datasets, numerical features are standardized and categorical features use either label-encoding or one-hot encoding; the choice of encoding has no appreciable effect on COLA’s performance. The train-test split is 7 : 3.

The optimality baseline in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)), which requires a Gurobi license to reproduce; open-source operations-research libraries such as Google-OR tools ([goo](#)) are a viable alternative. Solving the MILP in Appendix G is computationally expensive and is therefore practical only for small-scale datasets such as German Credit; computing the optimality baseline for other datasets requires reducing the number of features used.

We specify the model hyperparameters as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn models ([skl](#)) with all hyperparameters kept at their defaults. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses the linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interface with all hyperparameters at their defaults.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assist tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

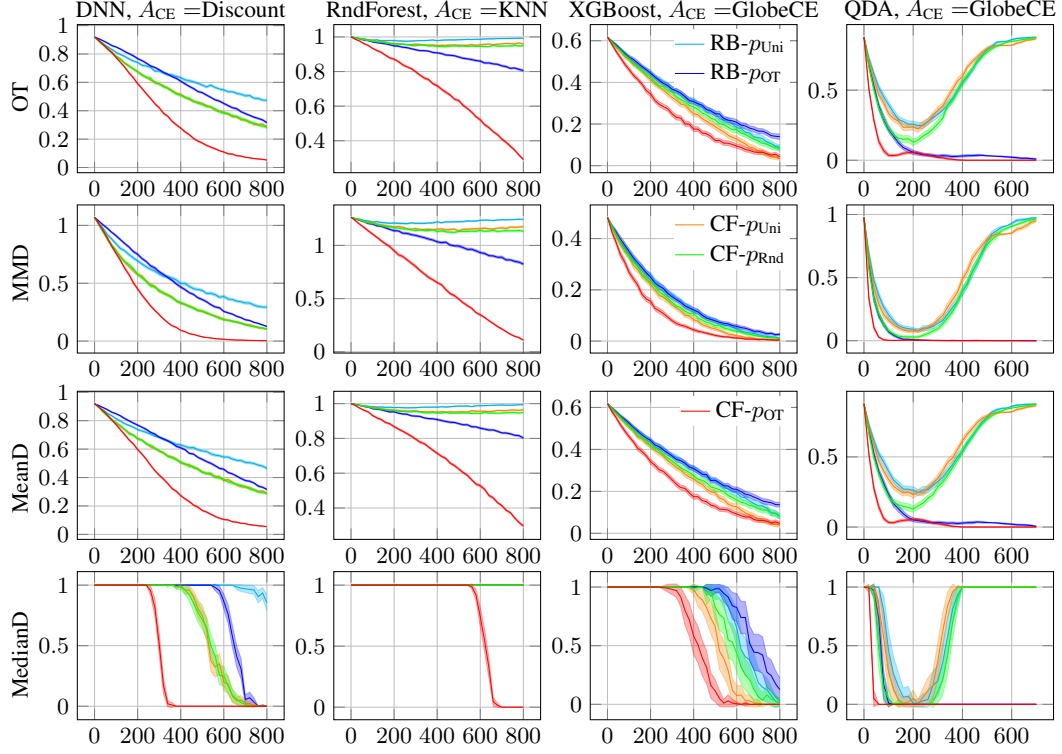


Figure 5: [HELOC] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , over 100 runs; shaded bands are 99.9% confidence intervals and the legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used. CF- p_{OT} (p -SHAP) reaches the target with the fewest actions throughout.

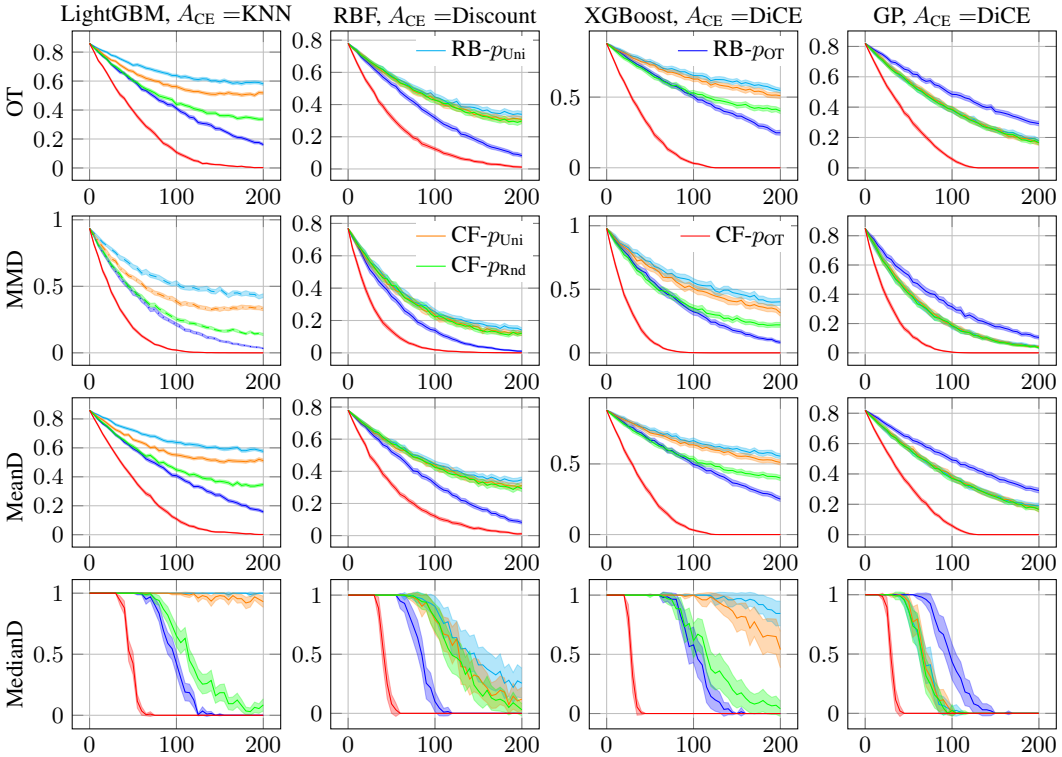


Figure 6: [German Credit] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , over 100 runs; shaded bands are 99.9% confidence intervals and the legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used. CF- p_{OT} (p -SHAP) again dominates the baselines.

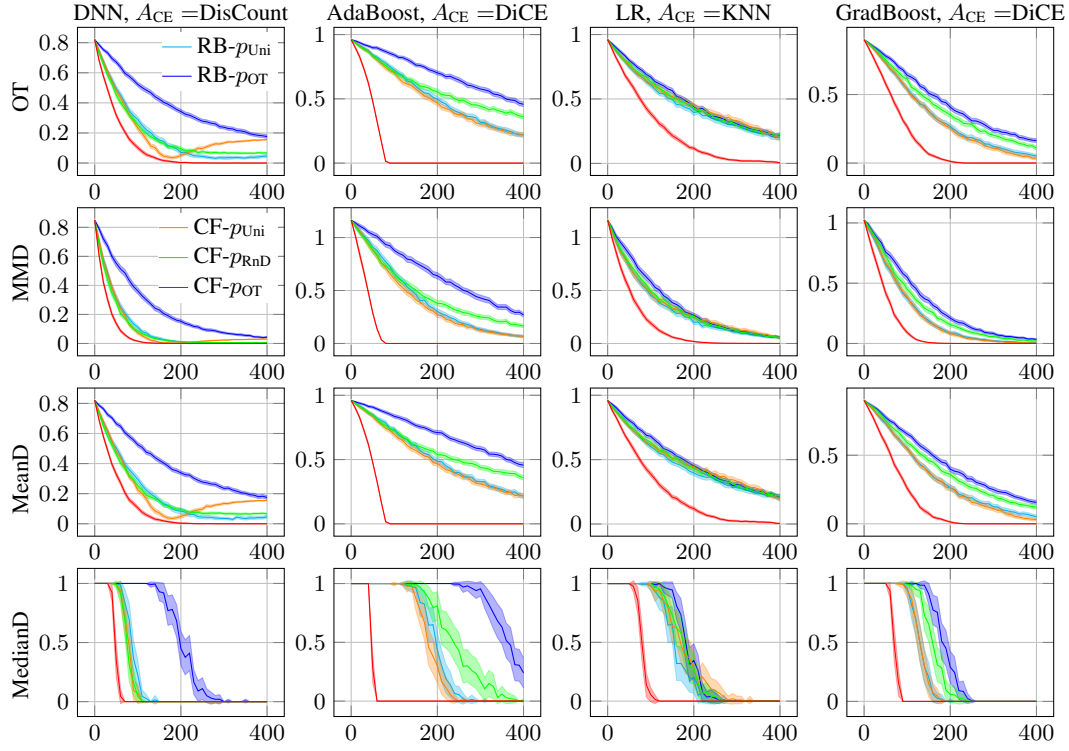


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , over 100 runs; shaded bands are 99.9% confidence intervals and the legends apply to all plots. $A_{\text{Value}}^{\max}$ is used. CF- p_{OT} (p -SHAP) consistently converges to the target fastest.

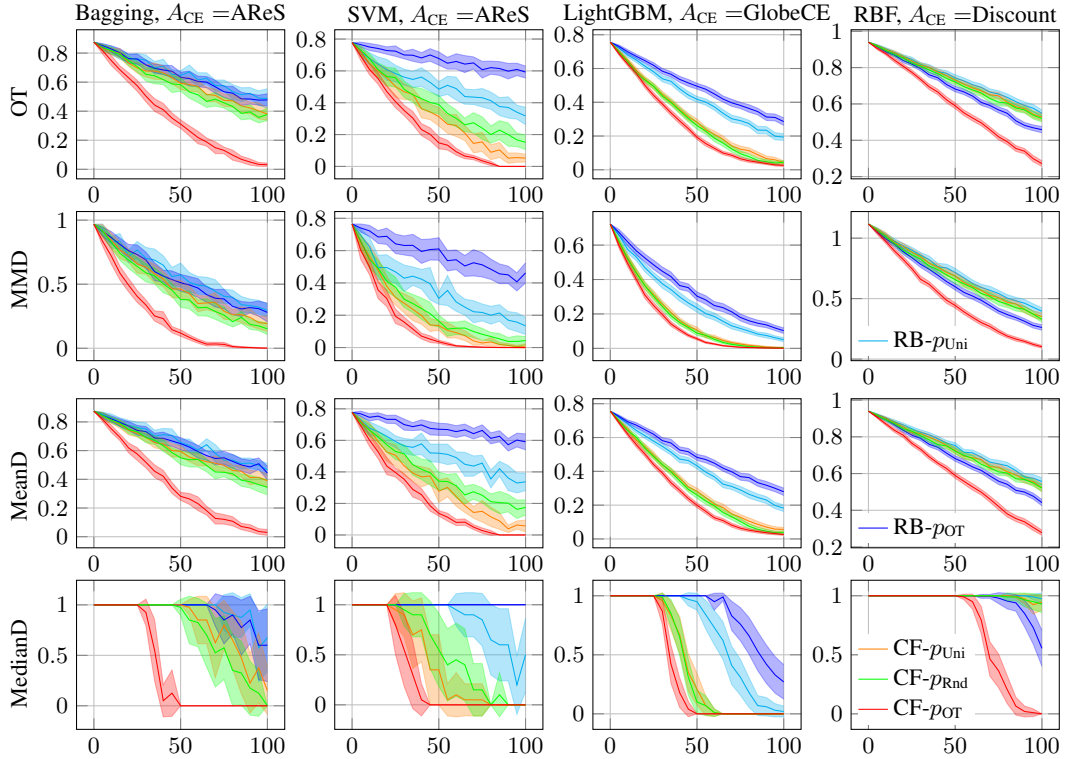


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , over 100 runs; shaded bands are 99.9% confidence intervals and the legends apply to all plots. $A_{\text{Value}}^{\max}$ is used. CF- p_{OT} (p -SHAP) retains its advantage across all models.